# Stochastic Traffic Signal Timing Optimization

A queuing and Monte Carlo study

**Dijith Dinesh**
**Athuljith S**
**Peter Joe Menachery**
**Pranav Manoj**

Department of Computer Engineering
Government Model Engineering College, Thrikkakara
Operations Research · PBCBT504 · Semester S5
October 26, 2026

# A precise, limited question

## Decision

Choose green durations for North, East, South, and West:

$$\mathbf{g} = (g_N, g_E, g_S, g_W), \qquad \sum_i g_i = 80 \text{ s}.$$

Each 96-second cycle includes 16 seconds of clearance.

## Fixed operation

Four protected phases; two-second service; turn shares of 20/60/20.

## Study case

Initially empty queues, assumed Poisson arrivals, 12 movement queues, and a 600-second horizon.

**Objective:** minimize expected cumulative waiting. No field measurements are used in this primary experiment.

# From arrivals to waiting cost

## Poisson arrivals

Uniform pseudorandom draws are transformed into exponential interarrival gaps:

$$\Delta t = -\frac{\ln(1-U)}{\lambda}.$$

A turning draw assigns each vehicle to a dedicated left, through, or right queue.

## Queue-area objective

$$D_T(\mathbf{g}) = \sum_{m=1}^{12} \int_0^T Q_m(t;\mathbf{g})\, dt.$$

This is cumulative waiting in vehicle-seconds, including partial waiting still in progress at the horizon.

# Search, then evaluate on fresh traffic

**1** Enumerate feasible timings

**2** Score on shared training arrivals

**3** Select the lowest estimated mean

**4** Evaluate on new, matched arrivals

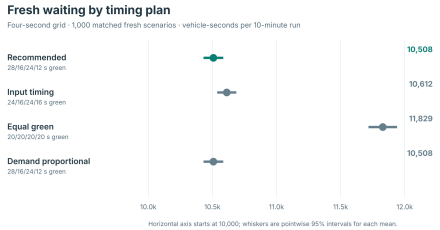
## Why simulation?

Service is gated by protected green phases and deterministic headways. A continuously serving M/M/1 formula does not represent this operation exactly.

## Reported measures

Mean waiting, paired differences, queue profiles, residual backlog, and uncertainty intervals.

# The ten-minute reference result




## Four-second-grid result

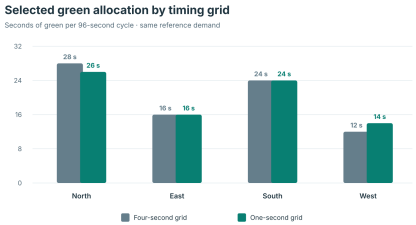
**Selected green (N/E/S/W)**
**28 / 16 / 24 / 12 s**

Fresh mean: **10,507.8** vehicle-seconds/run

Saving vs entered timing: **104.2** vehicle-seconds
1,000 fresh runs; paired 95% interval: 69.6–138.8.

The selected plan equals the demand-proportional baseline; the mean reduction is modest.

# Fine-grid timing result



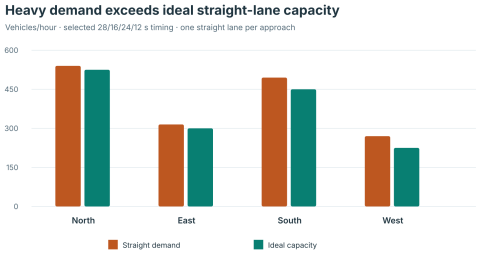

**26 / 16 / 24 / 14 s**

Selected green on the one-second grid

## Search and fresh evaluation

19,369 feasible allocations; 128 training runs each.
Search runtime: 6.86 seconds.
On 5,000 matched fresh runs: **44.8 vehicle-seconds/run** saved (0.42%).
Paired 95% interval: 34.8–54.8.

# Heavy demand can exceed modeled capacity



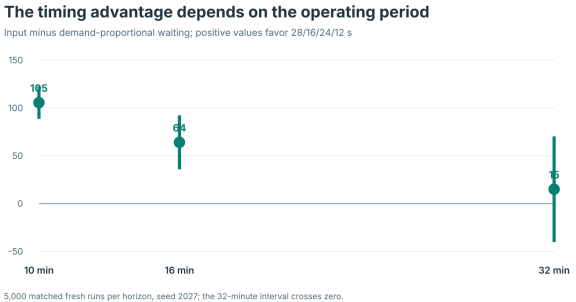

**2,700 veh/h**
Heavy-scenario total arrival rate

**86.4 s**
Green required by straight streams

**80 s**
Green available per cycle

Mean backlog: 54.4 vehicles at 10 min; 110.1 at 32 min.

# Interpret the recommendation within its scope




## 10 minutes

Saving: 105.5 vehicle-seconds per run
95% interval: 88.5–122.5

## 32 minutes

Saving: 15.0 vehicle-seconds per run
95% interval: $-40.3$–70.2; no resolved difference.

Field counts and calibration are needed before applying the timing outside this model.